

Comparison of Interquartile Ranges and Outlier Detection

Five-Number Summary

Class	Minimum	Q1	Median	Q3	Maximum
Day	32	56	74.5	82.5	99
Night	25.5	78	81	89	98

a. Comparison of Interquartile Ranges

The Interquartile Range (IQR) is calculated using the formula:

$$\text{IQR} = Q3 - Q1$$

Day Class

$$\text{IQR} = 82.5 - 56$$

$$\text{IQR} = 26.5$$

Night Class

$$\text{IQR} = 89 - 78$$

$$\text{IQR} = 11$$

b. Outlier Detection

Outliers are identified using the following ranges:

$$\text{Less Outlier Range} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Greater Outlier Range} = Q3 + 1.5 \times \text{IQR}$$

Day Class

$$\text{Less Outlier Range} = 56 - (1.5 \times 26.5)$$

$$= 56 - 39.75$$

$$= 16.25$$

$$\text{Greater Outlier Range} = 82.5 + (1.5 \times 26.5)$$

$$= 82.5 + 39.75$$

$$= 122.25$$

The minimum value (32) and maximum value (99) lie within the outlier ranges. Therefore, there are no outliers in the Day class.

Night Class

$$\text{Less Outlier Range} = 78 - (1.5 \times 11)$$

$$= 78 - 16.5$$

$$= 61.5$$

$$\text{Greater Outlier Range} = 89 + (1.5 \times 11)$$

$$= 89 + 16.5$$

$$= 105.5$$

The minimum value (25.5) is less than the less outlier Range (61.5). Therefore, 25.5 is identified as a less outlier. The maximum value (98) lies within the outlier ranges.

Conclusion

The Day class has a larger Interquartile Range (26.5) than the Night class (11), indicating greater variability in the Day class data. No outliers are present in the Day class. However, the Night class contains one lower outlier, with a value of 25.5.